

PUBLIC SUBMISSION

As of: March 21, 2025
Received: March 15, 2025
Status: [REDACTED]
Tracking No. m8b-1g9c-5yzl
Comments Due: March 15, 2025
Submission Type: Web

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-7749
Comment on FR Doc # 2025-02305

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General Comment

This whitepaper proposes the creation of an AI Benchmark to drive AI adoption in the US transportation industry.

Attachments

Transportation AI Benchmarking Whitepaper_20250315

TransportationBench: Driving Adoption through AI Benchmarking

Transportation agencies face increasing challenges in planning and operating a multimodal system that supports all models of travel. These include managing complex project development workflows across multiple stakeholders, understanding and responding to evolving travel patterns, identifying infrastructure gaps and opportunities, incorporating new mobility technologies across surface / air / maritime, and systematically capturing and applying lessons learned from implemented projects.

As a general-purpose technology, artificial intelligence (and generative AI, in particular) has shown promising potential to address these problems and is actively being adopted across sectors in the same way that electricity transformed society in the 20th century. Yet currently there is a lack of standardized validation methods essential for implementation and widespread AI adoption. A recent study¹ found that even leading AI models like Claude Sonnet 3.5 and GPT-4o demonstrate significant limitations when handling complex transportation concepts at the practitioner level. Transportation practitioners who have used models cite limitations in transportation domain knowledge and ‘hallucinations’ in model responses that hinders their usefulness beyond administrative tasks.

Trust is foundational to AI adoption and to sustain the adoption curve, we must work towards building transportation users trust in AI responses. While techniques such as retrieval augmented generation (RAG) help ground the AI responses in domain knowledge, our industry lacks the tools to systematically compare results of different large language models (LLMs) or their derivative applications. This results in individual public agencies validating AI solutions without an industry ‘benchmark’, leading to inconsistent validation approaches across jurisdictions, potentially redundant validation of the same AI solution(s), and overall inefficiencies that slows AI adoption.

A standardized AI benchmark would not only accelerate responsible AI adoption but also drive focused improvements in AI capabilities specifically relevant to transportation challenges. This ensures technological advances directly address the sector's most pressing needs – strengthening both our transportation system and the US position as the global leader in AI development.

This paper outlines an AI benchmark for transportation framework that: 1) identifies the current state of the practice for AI benchmarking and 2) proposes a validation approach for transportation knowledge – with the goal of enabling widespread AI adoption in transportation.

¹ University of Illinois. Benchmarking the Capabilities of Large Language Models in Transportation System Engineering: Accuracy, Consistency, and Reasoning Behaviors. Aug 2024.

Current State of the Practice for AI Benchmarking

AI benchmarks have grown in tandem with AI adoption and in theory, provide the means to objectively compare performance across math, sciences, and coding tests.

HuggingFace is a commonly used platform by the AI industry to stay up to date on model rankings for both open-source models (e.g. [Open LLM Leaderboard](#)) and proprietary models (e.g. [Chatbot Arena](#)). Rankings are based on model performance against several benchmarks as shown in **Figure 1** below and were developed by researchers from leading AI companies and universities.

Benchmark	Description
Graduate-Level Google-Proof Q&A (GPQA)	Evaluate models on 448 multi-choice questions on biology, physics, and chemistry. Tests models question-answering performance based on deep understanding and reasoning rather than fact recall or search.
Massive Multitask Language Understanding (MMLU)	Evaluate models on 57 tasks including elementary math, US history, computer science, law, etc. Tests model ability to demonstrate subject matter expertise, apply complex reasoning, and show consistent performance across domains.
MATH	Evaluate models on 12,500 competition math problems to test ability to apply mathematical principles, execute complex calculations and communicate answers clearly.
HumanEval	Evaluate models on 164 programming problems to test model ability to generate functionally correct and executable code.
SWE-Bench	Evaluate models on open-source repositories of real-world software bugs. Tests model ability to generate a software patch that resolves the issue without introducing new bugs.

Figure 1: Illustrative benchmarks used for LLM performance

In practice, however, there are limitations to these benchmarks. Evaluation researchers release datasets openly which can be sometimes integrated into the pre-training data for the LLMs. AI companies can ‘cherry-pick’ test cases where they have optimized their models to perform well in that also aligns with a specific benchmark. This results in rankings biased towards models that test well rather than provide an accurate indicator of performance. Most importantly, **these rankings tend to focus on LLM performance against academic benchmarks and are less relevant in understanding how LLMs or their derivative applications perform on domain specific knowledge.**

Recent efforts from the legal domain offer a potential framework for developing AI benchmarks for domain specific knowledge. LegalBench is an AI benchmark developed with data corpus and human expert review relating to the legal domain. Legal data corpus used includes private contracts, merger and acquisition documents of publicly traded companies, non-disclosure documents and privacy policies of consumer software applications. A team of experienced lawyers supervised 60,000+ annotations of the data corpus. **Figure 2** on the next page highlights LegalBench evaluation results of models’ ability to spot issues, recall relevant rules, and analyze their applications.

Data Corpus:  Contract Understanding Atticus Dataset (CUAD): Private Contracts  M&A Understanding Dataset (MAUD): M&A documents of public companies  ContractNLI: NDA related documents  Privacy QA: Privacy policies of consumer apps		Human Expert Annotation:  CUAD: 13K annotations labeled under supervision of experienced lawyers  MAUD: 47K annotations labeled under supervision of experienced lawyers	
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Model ↕	Cost In / Out ↕	Accuracy ▾	Latency (s) ↕
1  o1 Preview	\$15.00 / \$60.00	81.7 %	10.33 s
2  GPT 4o (2024-11-20)	\$2.50 / \$10.00	79.8 %	0.35 s
3  Qwen 2.5 Instruct Turbo (72B)	\$1.20 / \$1.20	79.2 %	0.62 s
4  Llama 3.1 Instruct Turbo (405B)	\$3.50 / \$3.50	79.0 %	0.81 s
5  GPT 4o (2024-08-06)	\$2.50 / \$10.00	79.0 %	0.39 s

Figure 2: LegalBench Data Corpus and Sample Evaluation of LLMs

AI Validation Approach for Transportation Knowledge

Drawing inspiration from LegalBench's success in evaluating AI performance in the legal domain, we propose **TransportationBench** – a specialized validation framework tailored to the transportation knowledge domain. This approach addresses the unique challenges identified in recent studies where even leading AI models demonstrate significant limitations when handling practitioner-level transportation concepts. This benchmark will establish an evaluation methodology built on three core elements: 1) General Transportation Data Corpus (GTDC); 2) Expert Annotation, Testing and Validation; 3) Benchmark Metrics. Details on each element are provided below and on the next page.

1. General Transportation Data Corpus: TransportationBench will utilize a curated data corpus of transportation technical standards and guidelines. Data corpora may include design guidelines such as the American Association of State Highway Transportation Officials (AASHTO) “Green Book” and/or test questions from common industry certifications exams such as the American Institute of Certified Planners (AICP). This corpus will serve as the foundation for evaluating AI model’s understanding of transportation concepts at the practitioner level. We will offer a public sample dataset to provide transparency while maintaining a larger, private dataset to protect test integrity and avoid leakage into foundation model training data.

2. Expert Annotation, Testing, Validation: Transportation domain experts representative of planning, engineering, operations and other relevant functions will supervise the document annotation, development of test cases, and validation. This process generally involves:

- a. **Annotation:** Experts systematically label data corpus and test cases that will serve as the “ground truth” for validation of AI responses. Annotations will identify key concepts, relationships, and acceptable answers for testing and evaluation. Test datasets will be used to train an LLM ‘Judge Model’.
- b. **Testing:** Experts craft test cases to challenge AI models on practitioner-level concepts and test performance in generating correct response across different tasks. Task formats may include simple question-answer (QA), multiple choice, general reasoning, large contexts, and multimodal (text, images, charts). Judge Model will apply test cases on subject LLMs to evaluate and provide capacity to supplement human evaluation at scale.
- c. **Validation:** Experts compare AI responses against expert-annotated ground truth using standardized metrics for accuracy, reasoning quality, and domain-specific appropriateness.

3. AI Benchmark Metrics:

- *Accuracy*: evaluates correctness of model outputs for each task and benchmark
- *Citation Quality*: gauges ability to reference appropriate standards and guidelines
- *Latency*: measures response time of models in returning a complete response
- *Cost*: analyzes the operational cost of running each model from an API provider

Figure 3 below illustrates example workflows of the proposed AI validation approach for transportation knowledge and can be scaled to accommodate new knowledge bases.

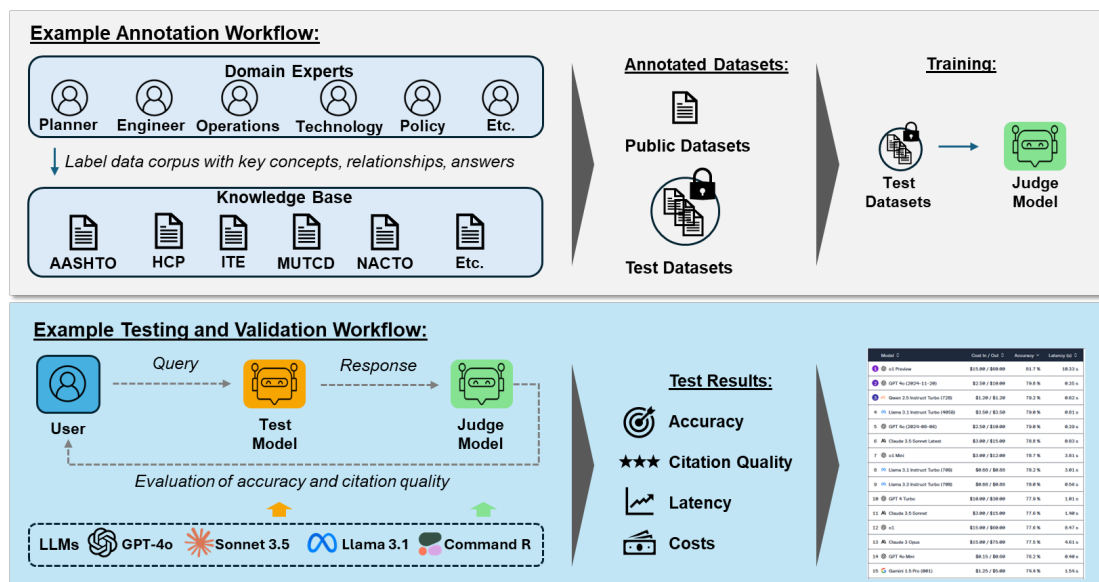


Figure 3: Illustrative Workflows for Annotation, Testing and Validation

Towards Widespread AI Adoption in Transportation

The TransportationBench framework represents a critical step toward responsible AI integration in the transportation sector. By establishing industry-specific benchmarks, systematic validation processes, and standardized metrics, we create a foundation that enables agencies to confidently evaluate and adopt AI technologies. This approach addresses current limitations while promoting consistency across jurisdictions, reducing redundant validation efforts, and accelerating implementation timelines. Most importantly, it builds user trust by ensuring AI responses on transportation concepts meets a practitioner level of standard. Through this collaborative framework, we can harness AI's transformative potential to address complex transportation challenges while maintaining the high standards our infrastructure systems demand.